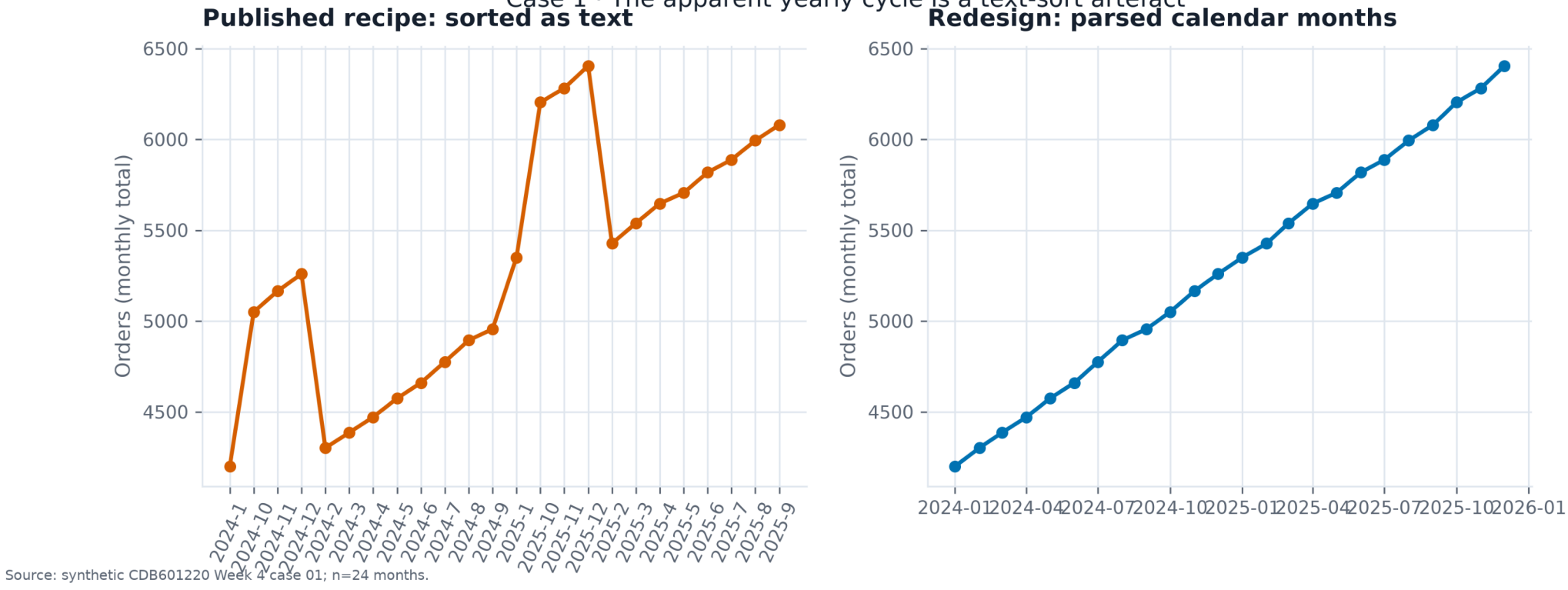
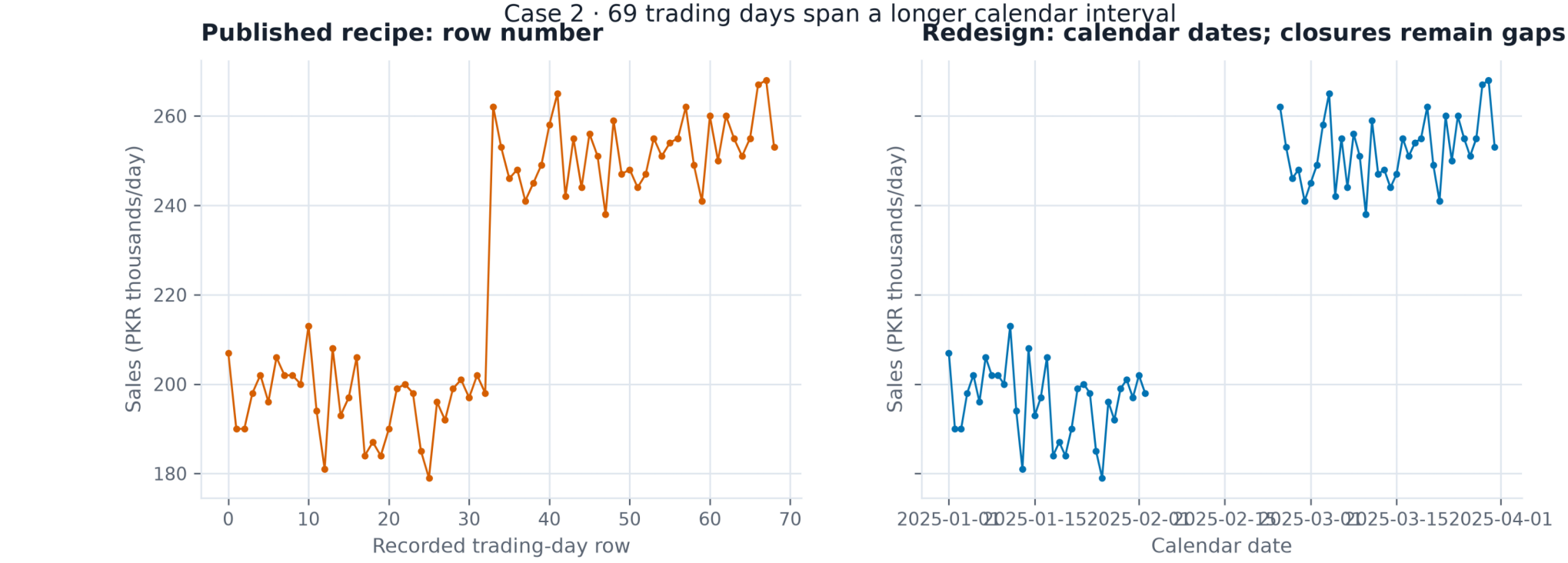


Case 1 — string-sorted months

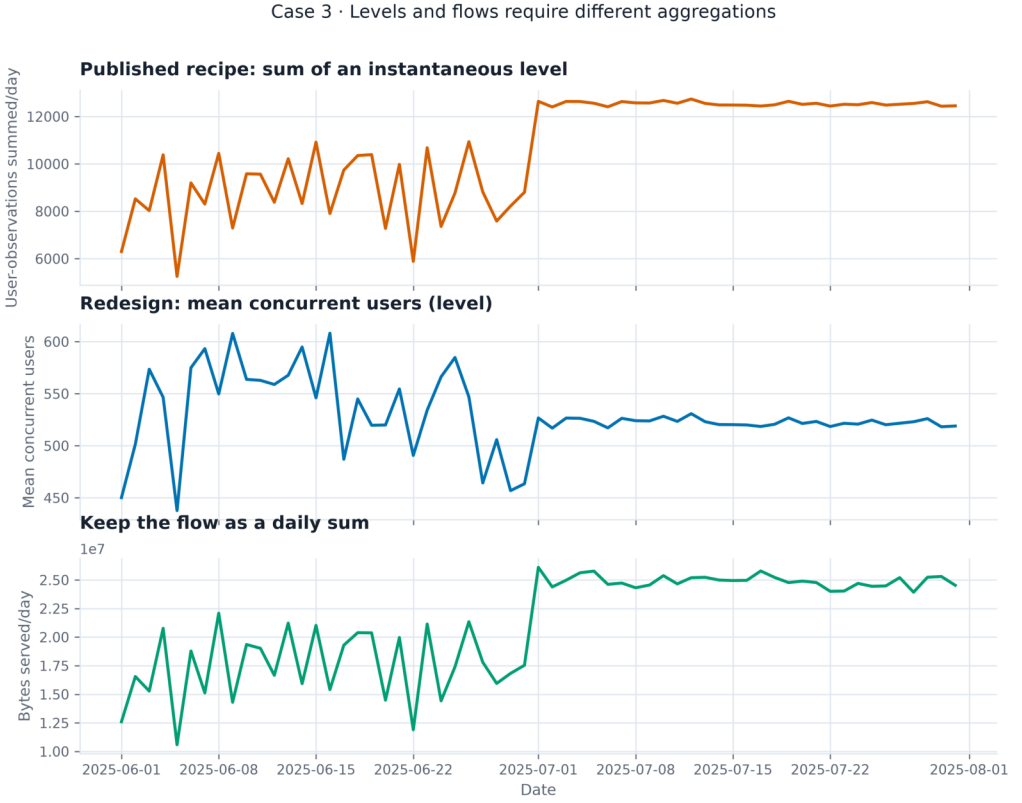
- Invited conclusion: Orders seem to rise and collapse in a repeating annual sequence.
- Decoy ruled out: The series is not too short: all 24 monthly observations are present.
- Mechanism: The YYYY-M strings sort lexicographically, so month 10 and 11 are placed before month 2; equal x spacing then assigns false time order.
- Evidence: 24 points reproduce the recipe's 958 and 978 drops; parsing YYYY-M chronologically changes their positions and removes the claimed repeated Q1 collapse.
- Redesign: Parse to a monthly DatetimeIndex, sort it, and draw the original values against calendar time.



Source: synthetic case 02; 69 recorded days; 21 unrecorded calendar dates.

Case 2 — row number hides closures

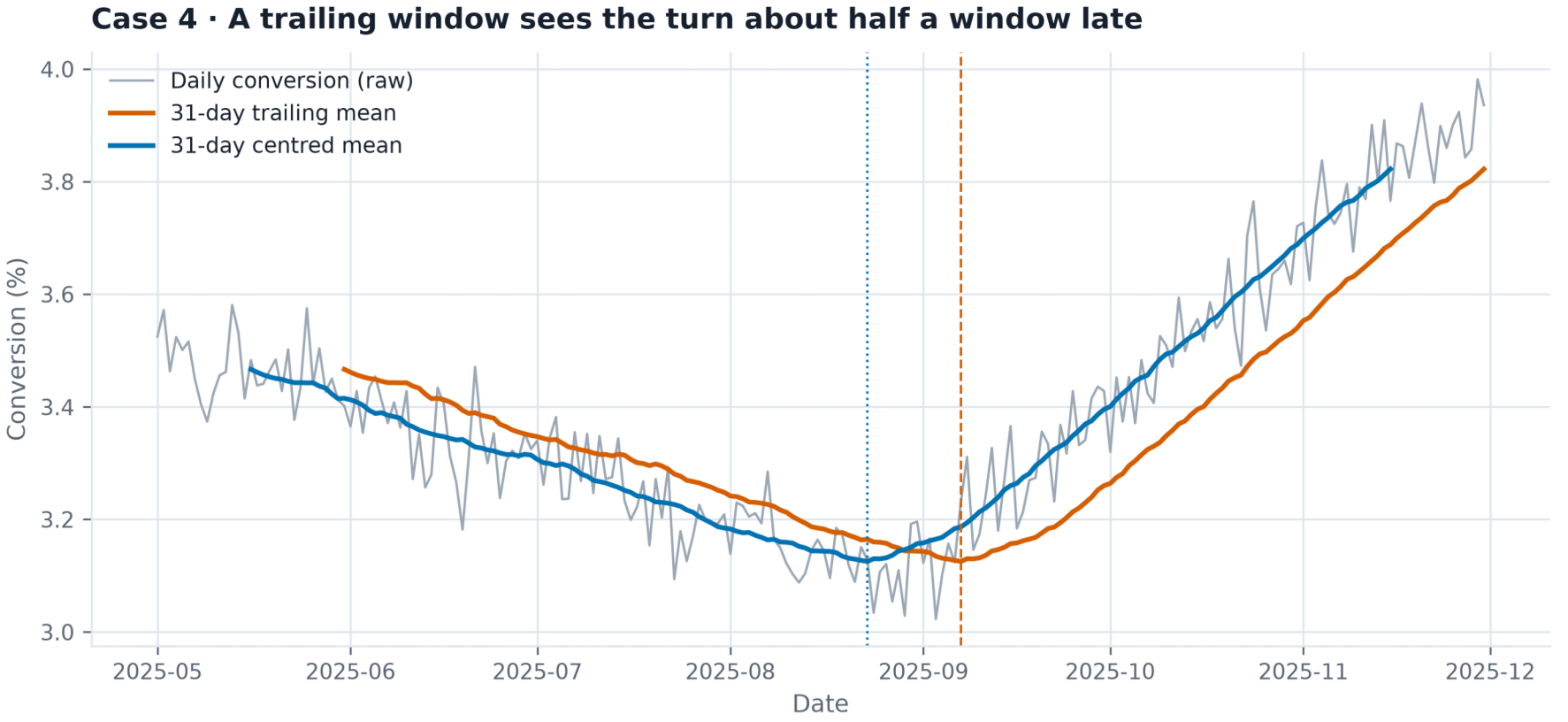
- Invited conclusion: Sales appear to rise steadily by about 22% across the quarter.
- Decoy ruled out: The endpoint arithmetic and 69 observations are correct; sample size is not the defect.
- Mechanism: Plotting row number compresses elapsed time by deleting non-trading calendar days from the x scale; it cannot distinguish a one-day step from a multi-day closure.
- Evidence: The 69 records span 90 calendar days with 21 unrecorded dates; the first-to-last change is +22.2%.
- Redesign: Plot the actual dates and reindex to daily frequency so closures create visible gaps.



Source: synthetic case 03; level=concurrent users, flow=bytes served.

Case 3 — summing a level

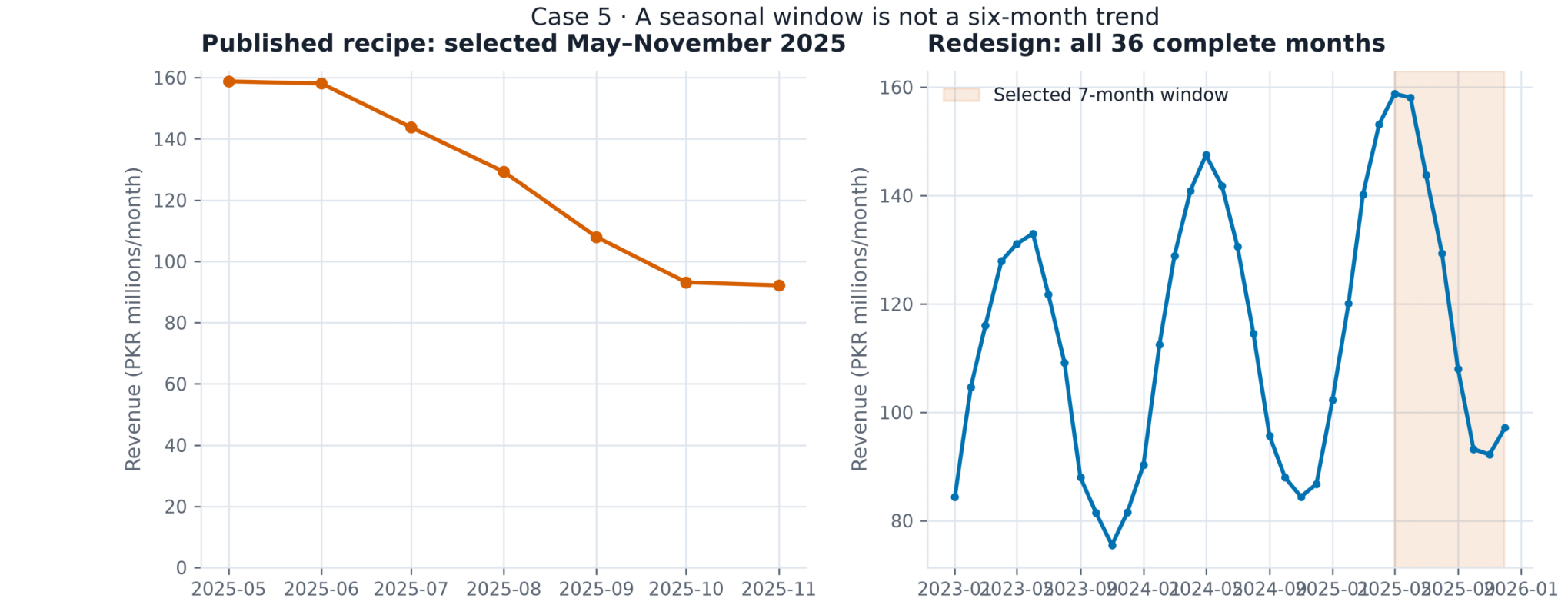
- Invited conclusion: Peak concurrent load appears to jump 43% overnight on 1 July.
- Decoy ruled out: The y-axis starts at zero and the June/July sums reproduce the recipe.
- Mechanism: Concurrent users is a level measured at an instant, not an additive flow; summing 24 snapshots gives a user-hours measure and compounds any change in sampling coverage.
- Evidence: The incorrect daily sums average 8,782 in June and 12,543 in July (+42.8%); daily means of the same level remove the multiplicative 24-hour sum, while bytes served remains a sum because it is a flow.
- Redesign: Aggregate concurrent users with a daily mean; aggregate bytes served with a daily sum, in separate labelled panels.



Source: synthetic case 04; 31-day windows; lag=15 days.

Case 4 — trailing-window lag

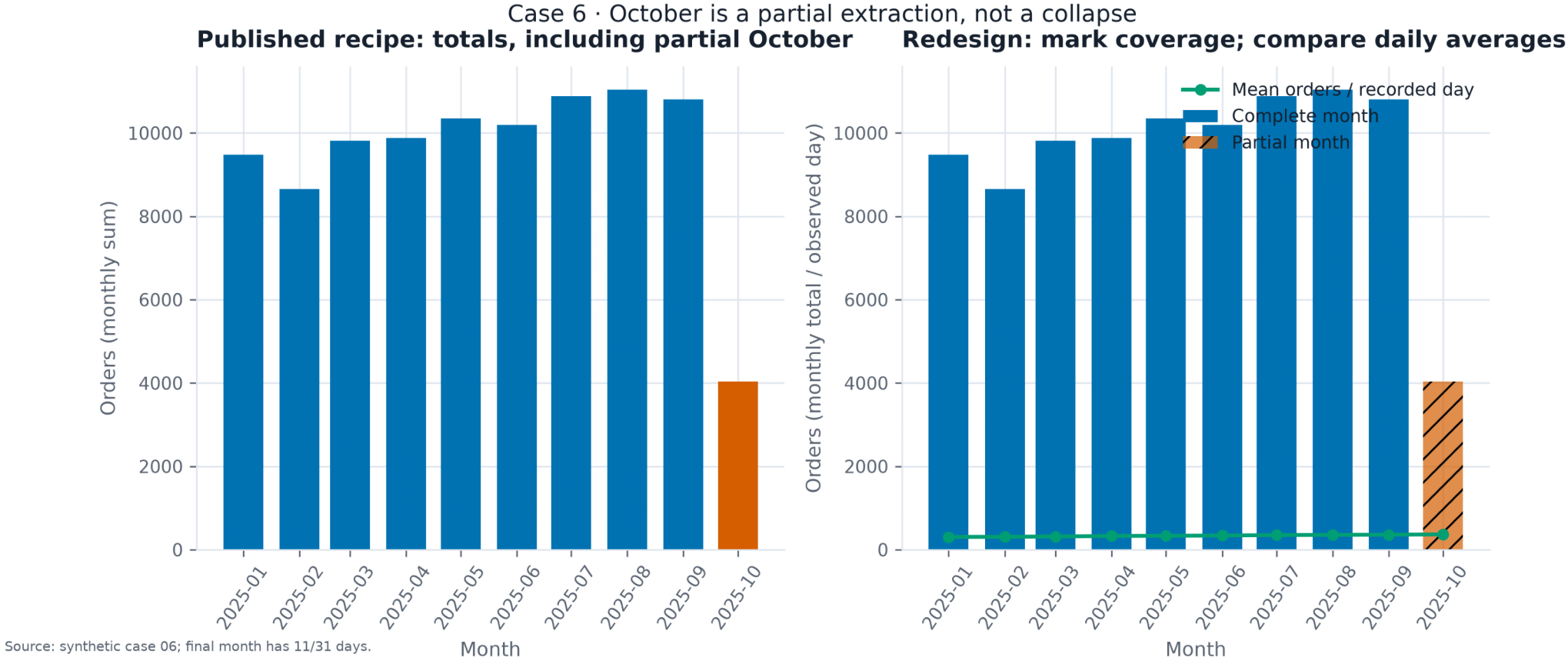
- Invited conclusion: The new checkout flow started working on 7 September.
- Decoy ruled out: The 31-day rolling mean is calculated on a sorted daily index, and its minimum is correctly 7 September.
- Mechanism: A trailing mean contains only current and earlier observations; at a 31-day window its minimum arrives roughly half a window after the centered estimate of the turn.
- Evidence: tslib.lag_at_turn reports trailing minimum 2025-09-07, centred minimum 2025-08-23, a lag of 15 days (half-window 15); the observational line alone does not establish that the checkout change caused a turn.
- Redesign: Keep the faint raw series, name the 31-day window, show the centred estimate for retrospective analysis, and label a trailing estimate as delayed on any live dashboard.



Source: synthetic case 05; 36 complete monthly observations; selected window is shown for context.

Case 5 — seasonally selected window

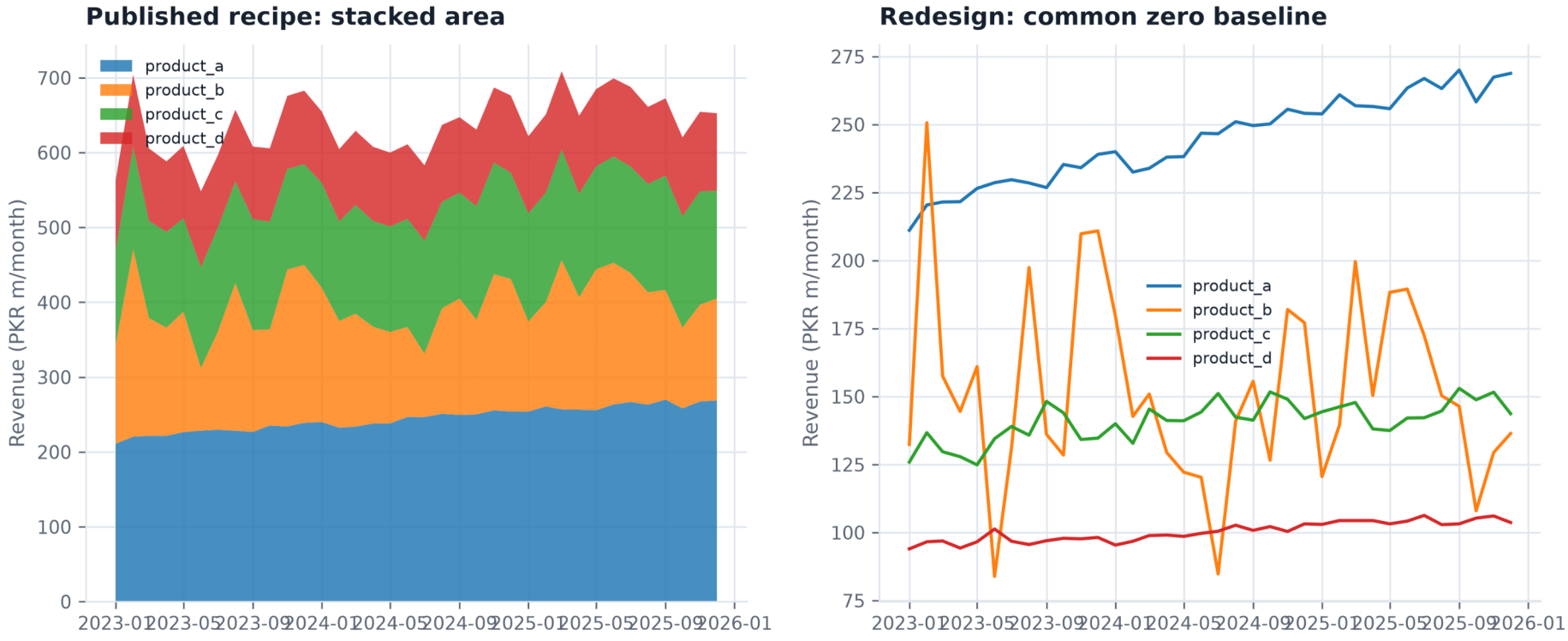
- Invited conclusion: Revenue has fallen 41.9% in six straight months.
- Decoy ruled out: The recipe uses seven points and an honest zero baseline; neither truncation nor n=7 explains it.
- Mechanism: The chosen window begins at a seasonal peak and ends near a trough; the crop selects phase of the annual cycle rather than estimating a sustained trend.
- Evidence: The chosen interval reproduces 158.8 to 92.2 PKR m (-41.9%), but the source has 36 complete months; the fall must be compared with the same months in other years.
- Redesign: Show all 36 months, identify the selected interval, and compare seasonal position across years instead of presenting the crop as a trend.



Case 6 — partial final month

- Invited conclusion: Orders collapsed in October: 4,041 versus September's 10,805.
- Decoy ruled out: Both bars start at zero and the monthly sum is arithmetically correct.
- Mechanism: October is only partially observed; comparing its subtotal with a complete September total confounds period coverage with the daily order rate.
- Evidence: October has 11 recorded days out of 31; its 4,041 orders are not a full-month total. The per-recorded-day mean is 367.4 versus 360.2 in September.
- Redesign: Hatch and label the partial month, and show daily average alongside full-period totals rather than treating an incomplete period as complete.

Case 7 · The top boundary is not Product D's value



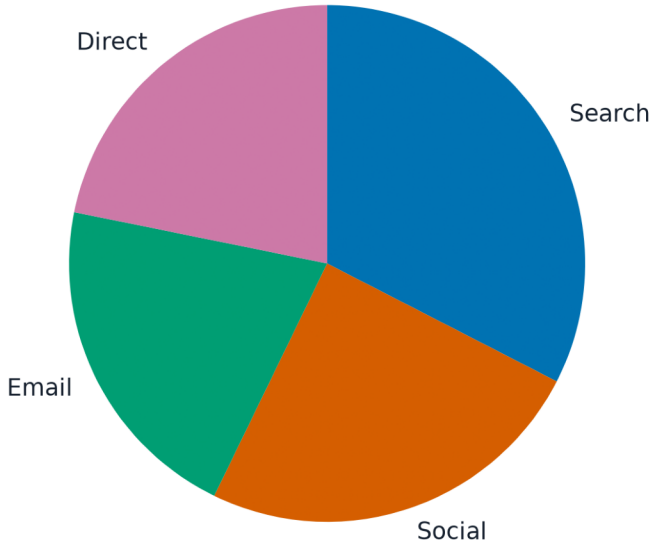
Source: synthetic case 07; stack total range 548.3–708.7 PKR m.

Case 7 — moving baseline in a stacked area

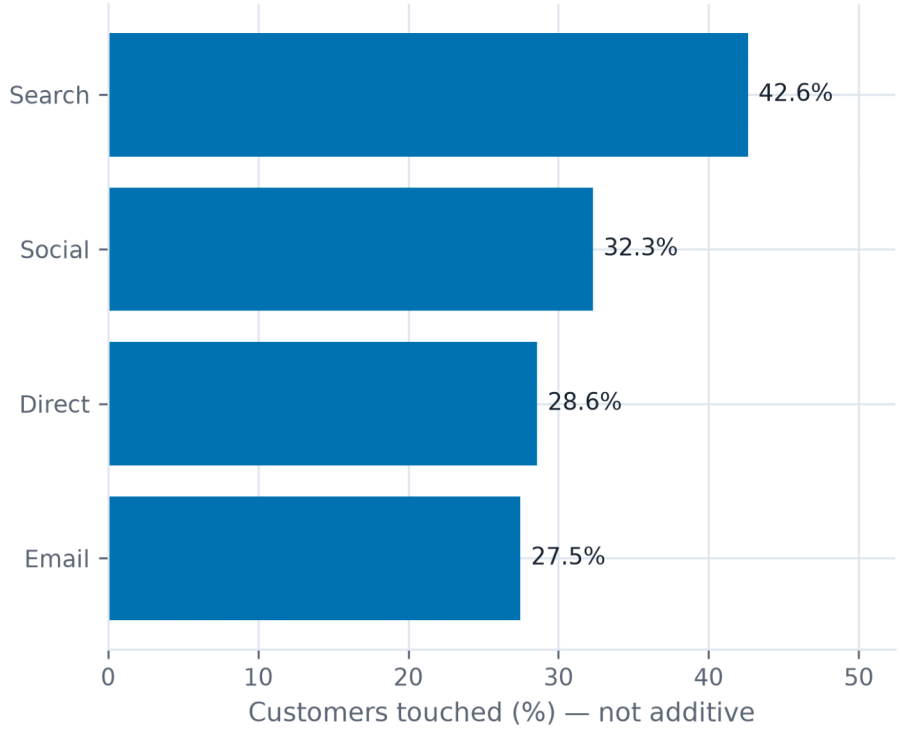
- Invited conclusion: Product D has been erratic since mid-2024.
- Decoy ruled out: The stack is legal: the four product values are exclusive and exhaustive, and the total envelope is not fabricated.
- Mechanism: The top layer's upper boundary is the cumulative sum of A+B+C+D; movement in the three lower layers shifts D's baseline, so its thickness—not its top edge—is its value.
- Evidence: The full stack ranges from 548.3 to 708.7 PKR m/month; in the original, D is read against a moving cumulative baseline.
- Redesign: Unstack the products into common-baseline lines (or small multiples) so D's own changes can be compared.

Case 8 · Four touch channels reach 131% of customers in total

Published recipe: pie implies exclusive parts



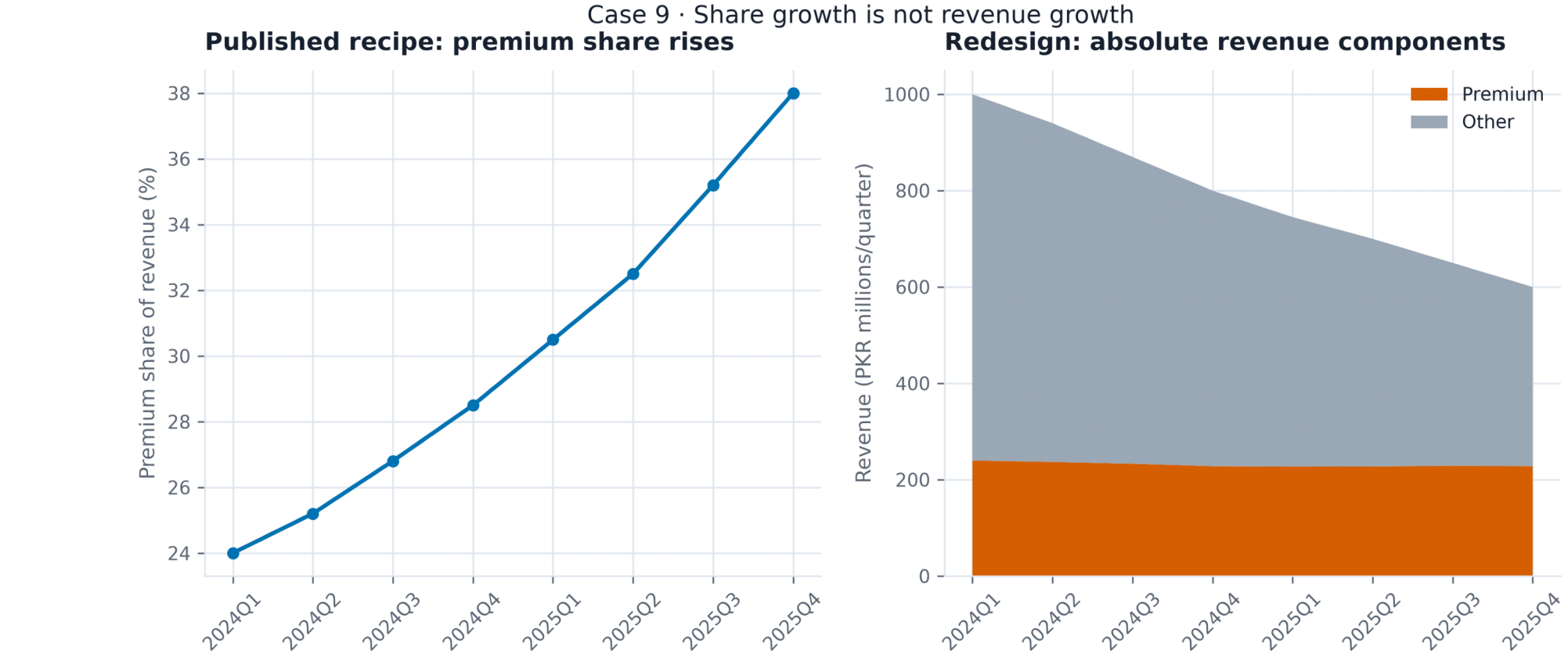
Redesign: overlapping customer reach



Source: synthetic case 08; 4,000 customers; 1,105 touched by multiple channels.

Case 8 — non-exclusive channel touches

- Invited conclusion: Paid search drives 42.6% of customers, apparently the largest share.
- Decoy ruled out: The row is one customer and each percentage correctly divides a touch count by 4,000.
- Mechanism: Attribution flags are not mutually exclusive: a customer may be touched by several channels, so the four marginal percentages do not partition a meaningful whole and a pie renormalizes them deceptively.
- Evidence: The channel rates are search 42.6%, social 32.3%, email 27.5%, direct 28.6%; they sum to 131.0%, and 1,105 customers have multiple true flags.
- Redesign: Use sorted bars labelled as percent of customers touched; explicitly state that overlaps mean bars need not sum to 100%.



Source: synthetic case 09; premium and other are exclusive components of the total.

Case 9 — a rising share of a changing total

- Invited conclusion: Premium is now 38% of revenue, a record share.
- Decoy ruled out: The share calculation and zero baseline are correct; the issue is not a percentage arithmetic error.
- Mechanism: The denominator (total revenue) is moving: a larger premium fraction does not establish that premium revenue itself grew or that overall revenue improved.
- Evidence: Premium share changes 24.0%→38.0%; total revenue changes 1,000.0→600.0 PKR m, and premium revenue changes 240.0→228.0 PKR m.
- Redesign: Keep the share, but place the exclusive premium/other revenue amounts beside it so the changing denominator is visible.